

# Evaluation setup

135

benchmark circuits

4–36

qubits per circuit

60 min

search budget · 3 trials

---

## Benchmarks and targets

QAOA, VQE, QPE, QFT,  
Grover and Shor families

### IBM-Eagle and Nam

15–17,438 input gates

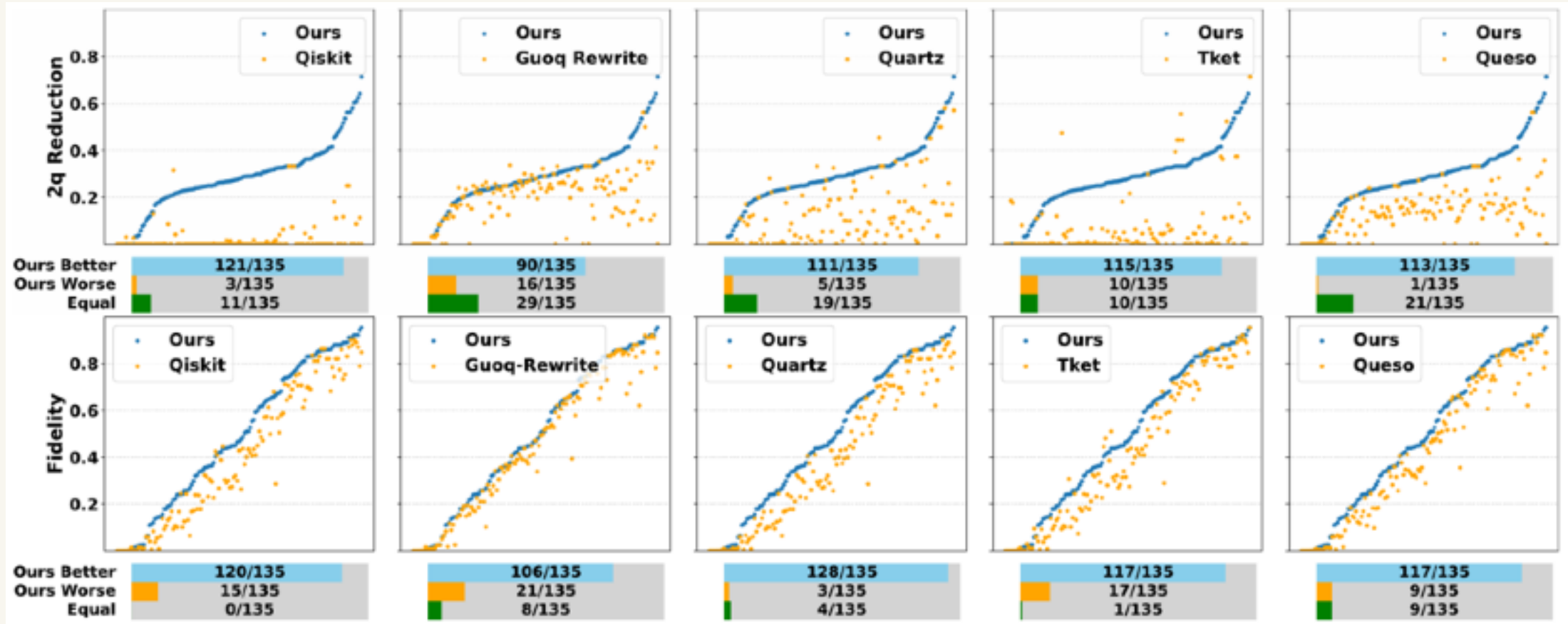
## QSymb search

Up to 9 equality-saturation iterations,  
alternating with symbolic rewrites

# Results

## IBM-Eagle · 135 benchmarks · two-qubit reduction and circuit fidelity on IBM devices

Blue: QSymb · Orange: comparator · Benchmarks sorted by QSymb for each metric; higher is better



Share of benchmarks (n = 135): **Strictly better** / **Equal** / **Worse**

|          |       |      |       |       |       |       |       |       |      |       |      |       |       |       |      |
|----------|-------|------|-------|-------|-------|-------|-------|-------|------|-------|------|-------|-------|-------|------|
| 2Q       | 89.6% | 8.1% | 2.2%  | 66.7% | 21.5% | 11.9% | 82.2% | 14.1% | 3.7% | 85.2% | 7.4% | 7.4%  | 83.7% | 15.6% | 0.7% |
| Fidelity | 88.9% | 0.0% | 11.1% | 78.5% | 5.9%  | 15.6% | 94.8% | 3.0%  | 2.2% | 86.7% | 0.7% | 12.6% | 86.7% | 6.7%  | 6.7% |